

Geonu Lee (이건우)

AI Research Engineer

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Work Experiences

Jul. 2024 - Present	Full-time – SNUAILAB <i>AI Research Engineer</i> • Led the end-to-end development and deployment of an AI-based smartphone cover glass micro-defect inspection system, achieving a False Negative Rate (FNR) below 2%. • Managed technical alignment and crisis resolution with a manufacturing client, establishing technical trust that successfully secured strategic corporate investment. • Trained deep learning models for a battery pole tracking system and integrated real-time pipelines into a C++ SDK, stabilizing edge deployment for a specialized inspection equipment manufacturer. • First-authored “Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection” (<i>Neurocomputing</i>, Under Minor Revision).
Oct. 2023 - Apr. 2024	Full-time – ALCHERA <i>AI Research Engineer, Anomaly Analysis Team</i> • Developed and deployed DETR-based object detection models for FireScout, a real-time vision-based wildfire detection solution, achieving a 20%p performance improvement over the legacy baseline. • Reconfigured the evaluation pipeline from image-level to instance-level metrics, resolving legacy evaluation limitations to establish a precise and robust performance measurement system.
Mar. 2023 - Sep. 2023	Intern – Naver Cloud <i>AI Research Engineer, Image Vision Team</i> • Contributed to the development of a face anti-spoofing service framework for eKYC services, constructing a robust system capable of detecting spoofing attacks across diverse environmental conditions. • First-authored a paper published at WACV 2025 on object anti-spoofing, improving HTER by 7.5% and EER by 6.6% to significantly enhance domain generalization performance.
Jan. 2020 - Feb. 2020	Intern – Electronics and Telecommunications Research Institute (ETRI) <i>AI Research Engineer</i> • Developed a human action recognition system that models interactions between humans and surrounding objects in video data.

Research Interests

Deep Learning, Pose Estimation, Multi-Modal, Vision-Language-Model, Vision-Language-Action, Anomaly Detection, Domain Generalization, Face Anti-Spoofing, Human Action Recognition, Human-Object Interaction Detection

Publications

- Jun. 2026 | **Geonu Lee**, Yujeong Oh, Geonhui Jang, Soyoung Lee, Jeonghyo Song, Sungmin Cha, YoungJoon Yoo, "Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection," in *Neurocomputing* (Minor Revision)
- Feb. 2025 | **Geonu Lee**, Yonghyun Jeong, Haneol Jang, YoungJoon Yoo, "Domain-Generalized Object Anti-Spoofing: Bridging Gaps and Patch Selection for Robust Detection across Domains," in *Winter Conference on Applications of Computer Vision (WACV)*
- Sep. 2022 | **Geonu Lee**, Kimin Yun, Jungchan Cho, "Occluded Pedestrian-Attribute Recognition for Video Sensors Using Group Sparsity," in *Sensors*, vol. 22, no. 17, pp. 6626
- Aug. 2022 | **Geonu Lee** and Jungchan Cho, "STDP-Net: Improved Pedestrian Attribute Recognition Using Swin Transformer and Semantic Self-Attention," in *IEEE ACCESS*, vol. 10, no.1, pp. 82656 - 82667
- Feb. 2021 | **Geonu Lee**, Kimin Yun, and Jungchan Cho, "Improved Human-Object Interaction Detection through On-the-Fly Stacked Generalization," in *IEEE ACCESS*, vol. 9, no. 1, pp. 34251-34263
- Feb. 2020 | Bhishan Bhandari, **Geonu Lee**, and Jungchan Cho, "Body-Part-Aware and Multitask-Aware Single-Image-Based Action Recognition," in *Applied Science*, vol. 10, no. 4, pp. 1531-1548

Research Projects

- 2022 | **A Study on the Complex Human Attributes for Situation Understanding**
Master's Research Project – Gachon University
- Conducted research on recognizing interactive human attributes in video sequences under complex contexts.
 - Utilized Swin Transformer and Transformer decoder architectures for semantic reasoning.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2021 | **A Study on the Understanding of Pedestrians**
Master's Research Project – Gachon University
- Developed a pedestrian attribute recognition method robust to occlusion.
 - Applied group sparsity regularization to handle various levels of visual obstruction.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2020 | **A Study on the Understanding of Human-Object-Interactions**
Undergraduate Research Project – Gachon University
- Designed a novel deep neural architecture based on on-the-fly stacked generalization for HOI detection.
 - Focused on modeling dynamic interactions between humans and surrounding objects.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2019 | **A Study on the Understanding of Human Situation Based on Deep Learning**
Undergraduate Research Project – Gachon University
- Proposed a multi-task learning framework combining human pose estimation and action recognition.
 - Addressed real-world situational understanding using contextual cues.
 - Funded by Electronics and Telecommunications Research Institute (ETRI).

Education

Mar. 2021 - Feb. 2023	Gachon University <i>M.S. in Software Engineering</i>
Mar. 2016 - Feb. 2021	Gachon University <i>B.S. in Department of Computer Engineering</i>